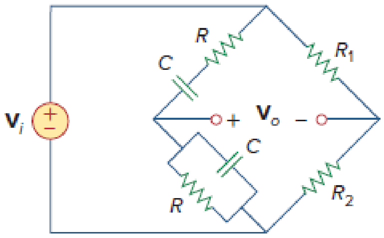
Chapter 10, Problem 90P

Problem

Figure 10.132 shows a Wien-bridge network. Show that the frequency at which the phase shift between the input and output signals is zero is $f=\frac{1}{2\pi}RC$, and that the necessary gain is **A**v = **V**o/**V**i = 3 at that frequency.

Figure 10.132:



Step-by-step solution

Step 1 of 9

Refer to Figure 10.132 in the textbook.

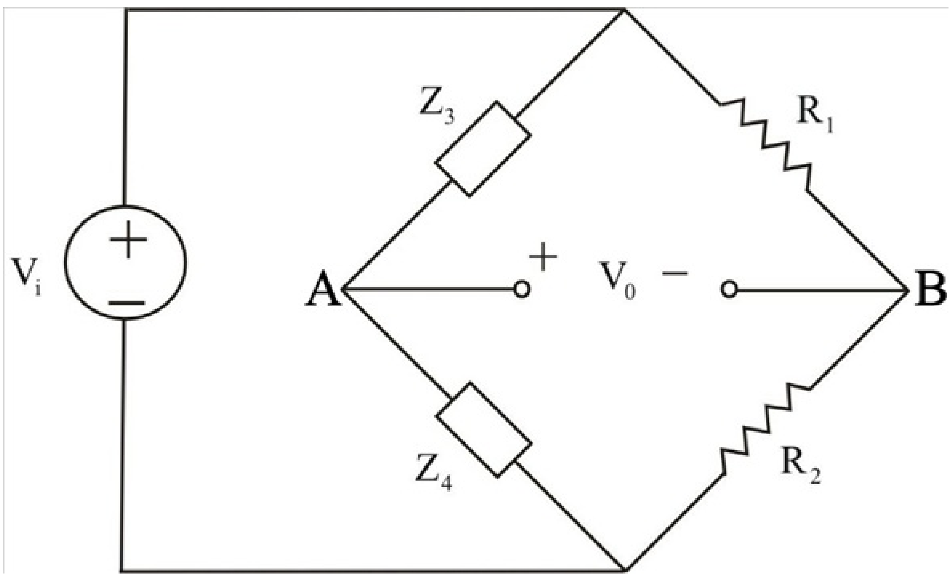
The following is the impedances of the bridge network.

$$
\begin{aligned}
Z_3 &= R+\frac{1}{j\omega C} \\
&= \frac{1+j\omega RC}{j\omega C}
\end{aligned}
$$

$$
\begin{aligned}
Z_4 &= R\|\frac{1}{j\omega C} \\
&= \frac{R}{1+j\omega RC}
\end{aligned}
$$

Step 2 of 9

Consider the circuit as shown in Figure 1.



Step 3 of 9

Figure 1

Step 4 of 9

Let V_A, and V_B be the voltage at point A, and B.

Determine the voltage at point A, V_A.

Apply voltage division rule in Figure 1.

$$V_A = \frac{Z_4}{Z_3 + Z_4} V_i$$

Determine the voltage at point B, V_B.

Apply voltage division rule in Figure 1.

$$V_B = \frac{R_2}{R_1 + R_2} V_i$$

Step 5 of 9

Determine the voltage V_o.

$$V_o = V_A - V_B$$

Substitute $\frac{Z_4}{Z_3 + Z_4} V_i$ for V_A, and $\frac{R_2}{R_1 + R_2} V_i$ for V_B.

$$V_0 = \left(\frac{Z_4}{Z_3 + Z_4} V_i \right) - \left(\frac{R_2}{R_1 + R_2} V_i \right)$$

$$V_0 = \left(\frac{Z_4}{Z_3 + Z_4} - \frac{R_2}{R_1 + R_2} \right) V_i$$

$$\frac{V_0}{V_i} = \left(\frac{Z_4}{Z_3 + Z_4} - \frac{R_2}{R_1 + R_2} \right)$$

Step 6 of 9

Substitute $\frac{1 + j\omega RC}{j\omega C}$ for Z_3, and $\frac{R}{1 + j\omega RC}$ for Z_4.

$$\frac{V_0}{V_i} = \frac{\left(\frac{R}{1 + j\omega RC} \right)}{\left(\frac{1 + j\omega RC}{j\omega C} \right) + \left(\frac{R}{1 + j\omega RC} \right)} - \left(\frac{R_2}{R_1 + R_2} \right)$$

$$= \frac{j\omega RC}{(1 + j\omega RC)^2 + j\omega RC} - \frac{R_2}{R_1 + R_2}$$

$$\frac{V_0}{V_i} = \frac{j\omega RC}{1 - \omega^2 R^2 C^2 + j3\omega RC} - \frac{R_2}{R_1 + R_2} \quad \ldots\ldots (1)$$

Step 7 of 9

For zero phase shift between the input V_i, and output V_0, then $\frac{V_0}{V_i}$ must be purely real.

$\frac{V_0}{V_i}$ will be purely real if $(1 - \omega^2 R^2 C^2 = 0)$ in equation (1).

$$1 - \omega^2 R^2 C^2 = 0$$
$$\omega^2 R^2 C^2 = 1$$
$$\omega = \frac{1}{RC}$$

Substitute $2\pi f$ for ω.

$$2\pi f = \frac{1}{RC}$$
$$f = \frac{1}{2\pi RC}$$

Step 8 of 9

Hence, the frequency at which the phase shift between the input and output signals is zero is $\boxed{f = \frac{1}{2\pi RC}}$.

Step 9 of 9

Determine the voltage gain A_v at frequency $f = \frac{1}{2\pi RC}$.

$$A_v = \frac{V_o}{V_i}$$

Substitute $\frac{j\omega RC}{1-\omega^2 R^2 C^2 + j3\omega RC}$ for $\frac{V_o}{V_i}$.

$$A_v = \frac{j\omega RC}{1-\omega^2 R^2 C^2 + j3\omega RC}$$

Substitute 0 for $(1 - \omega^2 R^2 C^2)$.

$$A_v = \frac{j\omega RC}{j3\omega RC}$$
$$= \frac{1}{3}$$

Therefore, the voltage gain A_v, at the frequency *f*, is $\boxed{\frac{1}{3}}$.